Lecture 13

- Optimization based estimation
- Factoring trajectories
- Factoring SLAM graph

Optimization Based Estimation

consider posterior:

$$p(x|y)$$

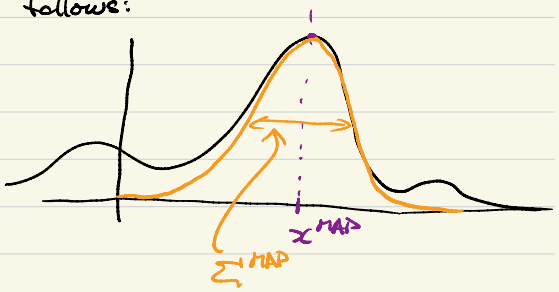
- suppose we can't find $p(x|y)$ explicitly.
- Can we find:

$$x^{MAP} = \arg\max_x p(x|y)$$

Maximum A Posteriori (MAP) Estimate

claim: since $p(x|y) = \frac{p(y|x)p(x)}{p(y)}$ we can sometimes found x^{MAP} even if we cannot find $p(x|y)$ in closed form.

In fact we can find a Gaussian approximation of $p(x|y)$ as follows:



the Gaussian $\mathcal{N}(x^{MAP}, \Sigma^{MAP})$ is called Laplace Approximation.

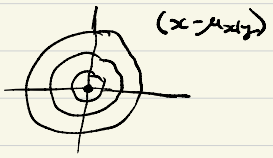
Let's start with Gaussian case to build intuition:

$$p(x|y) = \eta_{x|y}\, e^{-\frac{1}{2}(x-\mu_{x|y})^T \Sigma_{x|y}^{-1} (x-\mu_{x|y})}$$

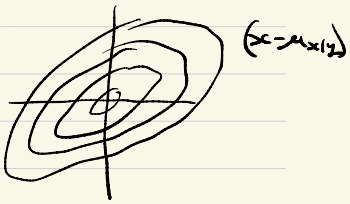
Notation: $\|x-\mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}} := (x-\mu_{x|y})^T \Sigma^{-1}_{x|y} (x-\mu_{x|y})$

(aside: Called Mahalanobis distance (squared))

Euclidean: or l_2 distance



Mahalanobis distance:



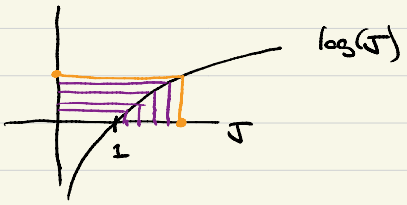
Let's solve:

$$x^{MAP} = \arg\max_x \eta_{x|y} e^{-\frac{1}{2}\|x-\mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}}}$$

"Negative Log Trick"

$$\arg\max_x \log p(x|y) = \arg\max_x p(x|y)$$

because $\log(\cdot)$ is monotonically increasing function



also $\arg\max_x J(x) = \arg\min_x -J(x)$

so we have:

$$\arg\min_x -\log(p(x|y)) = \arg\max_x p(x|y)$$

for $p(x|y)$ Gaussian we have:

$$-\log\left(\eta_{x|y} e^{-\frac{1}{2}\|x-\mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}}}\right)$$

$$= -\log \eta_{x|y} + \frac{1}{2}\|x-\mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}}$$

($-\log \eta_{x|y}$: constant)

$\Rightarrow$ does not change argmin.

Least Squares Optimization

$$x^{MAP} = \arg\min_x \frac{1}{2}\|x-\mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}}$$

convex quadratic optimization!

closed form solution!

find x^{MAP} with

$$\frac{\partial}{\partial x}\frac{1}{2}\|x-\mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}}\Big|_{x=x^{MAP}} = 0$$

$$\Sigma^{-1}_{x|y}(x-\mu_{x|y})\Big|_{x=x^{MAP}} = 0$$

$$x^{MAP} - \mu_{x|y} = 0$$

$$x^{MAP} = \mu_{x|y}$$

notice for Gaussian: $\mu_{x|y} = x^{mode} = x^{MAP}$

Q: How to find covariance Σ^{MAP}?

A: Consider Hessian:

$$H(x) := \frac{\partial^2}{\partial x^2} - \log(p(x|y))$$

for Gaussian:

$$\frac{\partial}{\partial x} - \log(p(x|y)) = \Sigma^{-1}_{x|y}(x-\mu_{x|y})$$

so

$$\frac{\partial^2}{\partial x^2} - \log(p(x|y)) = \frac{\partial}{\partial x}\Sigma^{-1}_{x|y}(x-\mu_{x|y})$$

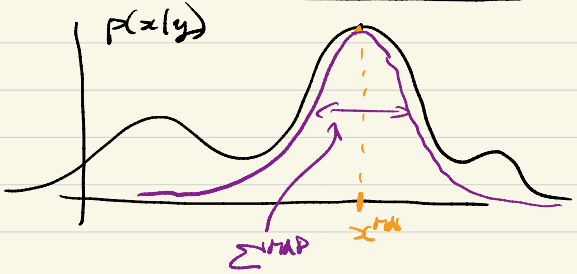
$$= \Sigma^{-1}_{x|y}$$

so

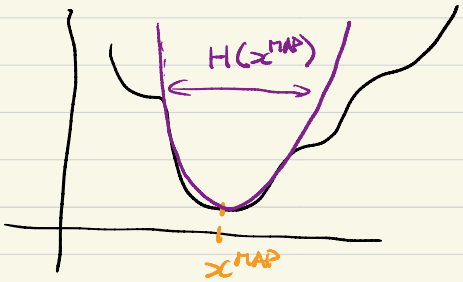
$$\boxed{\Sigma^{-1}_{x|y} = H(x^{MAP})^{-1}}$$

In general Laplace approximation:

$$\boxed{\begin{array}{c}\mathcal{N}(x^{MAP}, \Sigma^{MAP}) \\ \text{where } \Sigma^{MAP} = H(x^{MAP})^{-1}\end{array}}$$



$\Downarrow -\log p(x|y)$



Consider now Bayes' Rule:

$$p(x|y) = \frac{p(y|x)\,p(x)}{p(y)}$$

$$-\log(p(x|y)) = -\log(p(y|x)) - \log(p(x)) + \log(p(y))$$

constant in x so we can ignore

consider Linear Gaussian Model:

$Y = CX + V, \quad V \sim N(0, R)$

we need probability density:

$p(y|x)$

$Y - CX = V \sim \eta_v e^{-\frac{1}{2}\|v\|^2_{R^{-1}}}$

so $(Y - CX) \sim \eta e^{-\frac{1}{2}\|y - Cx\|^2_{R^{-1}}}$

$= p(y|x)$

$-\log(p(y|x)) = -\log \eta$ (can ignore) $+ \frac{1}{2}\|y - Cx\|^2_{R^{-1}}$

prior: $x \sim N(\mu_x, \Sigma_x)$

$-\log p(x) = -\log \eta_x + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$

ignore constant

We get x^{MAP} as follows:

$$x^{MAP} = \underset{x}{\operatorname{argmin}} \left(\frac{1}{2}\|y - Cx\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}} \right)$$

claim:

$x^{MAP} = \mu_{k+1}$ from KF update

$\Sigma^{MAP} = H(x^{MAP})^{-1}$ is given by KF covariance update Σ_{k+1}

(you show in PSET!)

Now consider nonlinear measurement model:

$Y = g(X) + V, \quad V \sim N(0, R)$

$p(y|x) = \eta e^{-\frac{1}{2}\|y - g(x)\|^2_{R^{-1}}}$

note: not Gaussian in x

$-\log(p(y|x)) = -\log(\eta)$ (ignore) $+ \frac{1}{2}\|y - g(x)\|^2_{R^{-1}}$

Nonlinear least squares

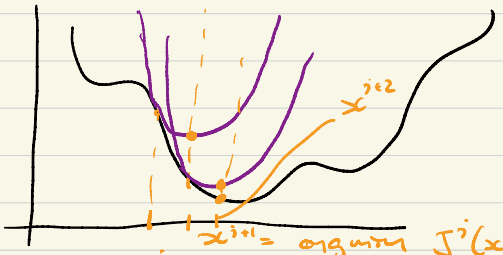
$$x^{MAP} = \underset{x}{\operatorname{argmin}} \left(\frac{1}{2}\|y - g(x)\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}} \right)$$

$\Sigma^{MAP} = H(x^{MAP})^{-1}$

Q: Is this just μ_{k+1}, Σ_{k+1} from EKF?

A: No, but related!

Gauss-Newton Algorithm (for optimizing Nonlinear Least Squares)



↳ $J^i(x)$ = 2nd order Taylor Approx of J centered at x^i

In Gauss-Newton we approximate $J^i(x)$ with:

$J(x) = \frac{1}{2}\|y - g(x)\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$

let $C(x^i) = \left.\frac{\partial g(x)}{\partial x}\right|_{x = x^i}$

$J^i(x) = \frac{1}{2}\|y - C(x^i)x\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$

$x^{i+1} = \underset{x}{\operatorname{argmin}} J^i(x)$

$J^{i+1} \Rightarrow$ plug in or linearize about x^{i+1}

⋮

claim: after 1 iteration we get the EKF estimate:

$x^{1+1} = x_{t|t}$ from EKF

$\Sigma^{1+1} = \Sigma_{t|t}$ from EKF

We can improve the EKF by continuing to iterate:

Iterated EKF (IEKF)

⇒ solves update step through Gauss-Newton iterations

Factoring the Trajectory

We can apply MAP estimation to a whole trajectory:

consider joint $(X_{0:t}, Y_{1:t}) \sim p(x_{0:t}, y_{1:t})$

recall: $p(x_{0:t}|y_{1:t}) = \frac{p(x_{0:t}, y_{1:t})}{p(y_{1:t})}$ (constant in x)

so, (posterior, joint)

$$x_{0:t}^{MAP} = \underset{x_{0:t}}{\operatorname{argmax}}\, p(x_{0:t}|y_{1:t}) = \underset{x_{0:t}}{\operatorname{argmax}}\, p(x_{0:t}, y_{1:t})$$

claim: we can always factor the traj. (assuming Cond. Indep. measurements and Markovianity) into

$$p(x_{0:t}, y_{1:t}) = \sum_{\tau=1}^{t}\Big(-\log(p(y_\tau|x_\tau)) - \log(p(x_\tau|x_{\tau-1}))\Big) + -\log(p(x_0))$$

proof:

$$p(x_{0:t}, y_{1:t}) = \underbrace{p(y_t|x_{0:t}, y_{1:t-1})}_{= p(y_t|x_t) \text{ Cond. Ind. measurements}} \times p(x_{0:t}, y_{1:t-1})$$

$$= p(y_t|x_t)\, \underbrace{p(x_t|x_{0:t-1}, y_{1:t-1})}_{= p(x_t|x_{t-1}) \text{ by Markovianity}} p(x_{0:t-1}, y_{1:t-1})$$

$$= p(y_t|x_t)p(x_t|x_{t-1})\, p(x_{0:t-1}, y_{1:t-1}) \quad \text{(recursion!)}$$

$$\vdots$$

$$= \prod_{\tau=1}^{t} p(y_\tau|x_\tau)\, p(x_\tau|x_{\tau-1})\; p(x_0)$$

taking $-\log(p(x_{0:t}, y_{1:t}))$ we get

$$\sum_{\tau=1}^{t}\Big(-\log(p(y_\tau|x_\tau)) - \log(p(x_\tau|x_{\tau-1}))\Big) - \log(p(x_0))$$

□

Consider nonlinear SS model with Gaussian white noise:

$$Y_t = g(X_t) + V_t,\quad V_t \sim GWN(0, R_t)$$

$$X_t = f(X_{t-1}, u_{t-1}) + W_{t-1},\quad W_{t-1} \sim GWN(0, Q_{t-1})$$

$$X_0 \sim N(\mu_{0|0}, \Sigma_{0|0})$$

and $V_{1:t}, W_{0:t}, X_0$ all mutually independent.

We get:

$$-\log(p(y_t|x_t)) = \frac{1}{2}\|y_t - g(x_t)\|^2_{R_t^{-1}} + \underbrace{Const}_{\text{ignore}}$$

$$-\log(p(x_t|x_{t-1})) = \frac{1}{2}\|x_t - f(x_{t-1}, u_{t-1})\|^2_{Q_{t-1}^{-1}} + \underbrace{Const}_{\text{ignore}}$$

$$-\log(p(x_0)) = \frac{1}{2}\|x_0 - \mu_{0|0}\|^2_{\Sigma_{0|0}^{-1}} + \underbrace{Const}_{\text{ignore}}$$

We get the full trajectory

$$x_{0:t}^{MAP} = \underset{x_{0:t}}{\operatorname{arg\,min}} \sum_{\tau=1}^{t}\Big(\frac{1}{2}\|y_\tau - g(x_\tau)\|^2_{R_\tau^{-1}} + \frac{1}{2}\|x_\tau - f(x_{\tau-1}, u_{\tau-1})\|^2_{Q_\tau^{-1}}\Big) + \frac{1}{2}\|x_0 - \mu_{0|0}\|^2_{\Sigma_{0|0}^{-1}}$$

Nonlinear least squares!

- Solve to obtain $x_{0:t}^{MAP}$, called "smoothed" trajectory estimate.
- Can be better than the filter estimate because x_τ^{MAP} uses all measurements $y_{1:t}$, not just $y_{1:\tau}$ (for $\tau < t$).